

[3]
✓ 1s

```
import pandas as pd
import matplotlib.pyplot as plt

GARIS = "=" * 60

def judul(teks):
    print("\n" + GARIS)
    print(f" {teks}")
    print(GARIS)

# A. PENGORGANISASIAN DATA

# Tabel Data (file excel satu folder dengan script ini)
path = "data.xlsx"
dataraw = pd.read_excel(path)

judul("A1. TABEL DATA MENTAH")
print(dataraw.to_string(index=False))

# Tabel Frekuensi
datafrq = pd.crosstab(index=dataraw["Grade"], columns="Frekuensi")
datafrq.columns.name = None

# Modifikasi: tambah persentase & total
tabel = datafrq.copy()
tabel["Persentase (%)"] = (tabel["Frekuensi"] / tabel["Frekuensi"].sum() * 100).round(1)
tabel.loc["Total"] = [tabel["Frekuensi"].sum(), 100.0]
tabel["Frekuensi"] = tabel["Frekuensi"].astype(int)

judul("A2. TABEL FREKUENSI GRADE")
print(tabel)

# B. PENYAJIAN DATA (GRAFIK)
```

[3]
✓ 1s



```
print(tabel)

# B. PENYAJIAN DATA (GRAFIK)

warna = ["#4C72B0", "#55A868", "#8172B2", "#64B5CD", "#CCB974", "#DD8452", "#C44E52"]

# Grafik Garis
ax = datafrq.plot(marker="o", color="#4C72B0", linewidth=2,
                  title="Grafik Garis Frekuensi Grade", grid=True)
ax.set_ylabel("Jumlah Mahasiswa")
ax.get_figure().savefig("grafik_garis.png", dpi=150, bbox_inches="tight")

# Grafik Batang
plt.figure()
ax = datafrq["Frekuensi"].plot(kind="bar", color=warna[:len(datafrq)], rot=0,
                              title="Grafik Batang Frekuensi Grade")
ax.set_ylabel("Jumlah Mahasiswa")
for p in ax.patches: # label angka di atas batang
    ax.annotate(int(p.get_height()), (p.get_x() + p.get_width() / 2, p.get_height()),
                ha="center", va="bottom")
ax.get_figure().savefig("grafik_batang.png", dpi=150, bbox_inches="tight")

# Pie Chart
plt.figure()
ax = datafrq.plot(kind="pie", y="Frekuensi", autopct="%1.0f%%", legend=False,
                  colors=warna[:len(datafrq)], startangle=90,
                  wedgeprops={"edgecolor": "white"}, pctdistance=0.78,
                  title="Pie Chart Persentase Grade")
ax.set_ylabel("")
ax.get_figure().savefig("pie_chart.png", dpi=150, bbox_inches="tight")

# C. STATISTIKA DESKRIPTIF-
dataraw["Nilai"] = pd.to_numeric(dataraw["Nilai"], errors="coerce")
dt = dataraw["Nilai"]
#dt = dt.describe()
```

[3]
✓ 1s



```
# C. STATISTIKA DESKRIPTIF-
dataraw["Nilai"] = pd.to_numeric(dataraw["Nilai"], errors="coerce")
dt = dataraw["Nilai"]
stats = dt.describe()
stats["Standard Error"] = dt.sem()
stats["Median"] = dt.median()
stats["Mode"] = dt.mode().iloc[0]
stats["variance"] = dt.var()
stats["range"] = dt.max() - dt.min()
stats["skewness"] = dt.skew()
stats["kurtosis"] = dt.kurtosis()

judul("C1. STATISTIKA DESKRIPTIF NILAI AKHIR")
print(stats.round(4).to_string())

# Modifikasi: analisis tambahan
q1, q3 = dt.quantile(0.25), dt.quantile(0.75)
iqr = q3 - q1
bawah, atas = q1 - 1.5 * iqr, q3 + 1.5 * iqr
outlier = dataraw[(dt < bawah) | (dt > atas)]
cv = dt.std() / dt.mean() * 100

judul("C2. ANALISIS TAMBAHAN")
print(f"Semua modus      : {' '.join(f'{m:.2f}' for m in dt.mode())}")
print(f"IQR (Q3 - Q1)    : {iqr:.2f}")
print(f"Batas bawah outlier : {bawah:.2f}")
print(f"Batas atas outlier  : {atas:.2f}")
print(f"Jumlah outlier       : {len(outlier)}")
print(f"Koefisien variasi    : {cv:.2f}%")
print(f"Mean vs Median       : {'Mean < Median -> menceng ke kiri' if dt.mean() < dt.median() else 'Mean >= Median'})

plt.show()
```

Output :

=====

A1. TABEL DATA MENTAH

=====

NIM	Name	KUIS	UTS	UAS	Nilai	Grade
120001	John Doe	85	78	90	84.44	A
120005	Jane Smith	92	88	94	91.38	A
120017	Michael Johnson	48	65	56	56.41	BC
120018	Emily Brown	90	85	88	87.65	A
120022	Daniel Martinez	75	80	77	77.35	AB
120024	Sarah Wilson	88	70	65	74.01	B
120025	David Garcia	75	70	75	73.35	B
120027	Jessica Lee	95	91	96	94.03	A
120034	Matthew Davis	79	84	78	80.30	AB
120035	Jennifer Rodriguez	42	48	63	51.33	C
120036	Andrew Taylor	53	64	59	58.73	BC
120038	Amanda Hernandez	91	95	90	91.97	A
120039	Christopher Clark	76	81	75	77.30	AB
120046	Elizabeth Martinez	89	93	88	89.97	A
120047	William Anderson	80	85	79	81.30	A
120049	Sophia Thomas	94	90	95	93.03	A
120054	James White	77	82	76	78.30	AB
120060	Olivia Taylor	86	89	84	86.29	A
120063	Benjamin Wilson	92	88	93	91.03	A
120070	Emma Garcia	78	83	77	79.30	AB
120071	Alexander Davis	41	53	44	46.01	C
120073	Mia Moore	68	49	54	56.83	BC
120082	Ethan Jackson	81	87	80	82.63	A
120083	Isabella Brown	87	90	85	87.29	A
120088	Noah Taylor	79	84	78	80.30	AB
120090	Abigail Johnson	70	77	64	70.21	B
120093	Logan Martinez	67	54	54	58.16	BC
120095	Avery Clark	69	62	71	67.39	B
120098	Jacob Thompson	76	81	75	77.30	AB
120099	Samantha Harris	60	65	64	63.05	BC

=====

✓ 1s

plt.snow()

▼

...

A2. TABEL FREKUENSI GRADE

	Frekuensi	Persentase (%)
Grade		
A	12	40.0
AB	7	23.3
B	4	13.3
BC	5	16.7
C	2	6.7
Total	30	100.0

C1. STATISTIKA DESKRIPTIF NILAI AKHIR

count	30.0000
mean	76.2213
std	13.4666
min	46.0100
25%	68.0950
50%	78.8000
75%	87.0400
max	94.0300
Standard Error	2.4587
Median	78.8000
Mode	77.3000
variance	181.3489
range	48.0200
skewness	-0.6642
kurtosis	-0.5275

C2. ANALISIS TAMBAHAN

Semua modus : 77.30, 80.30
IQR (Q3 - Q1) : 18.95
Batas bawah outlier : 39.68
Batas atas outlier : 115.46
Jumlah outlier : 0
Koefisien variasi : 17.67%
Mean vs Median : Mean < Median -> menceng ke kiri

Grafik Garis Frekuensi Grade



